

SAP CPI Project Documentation

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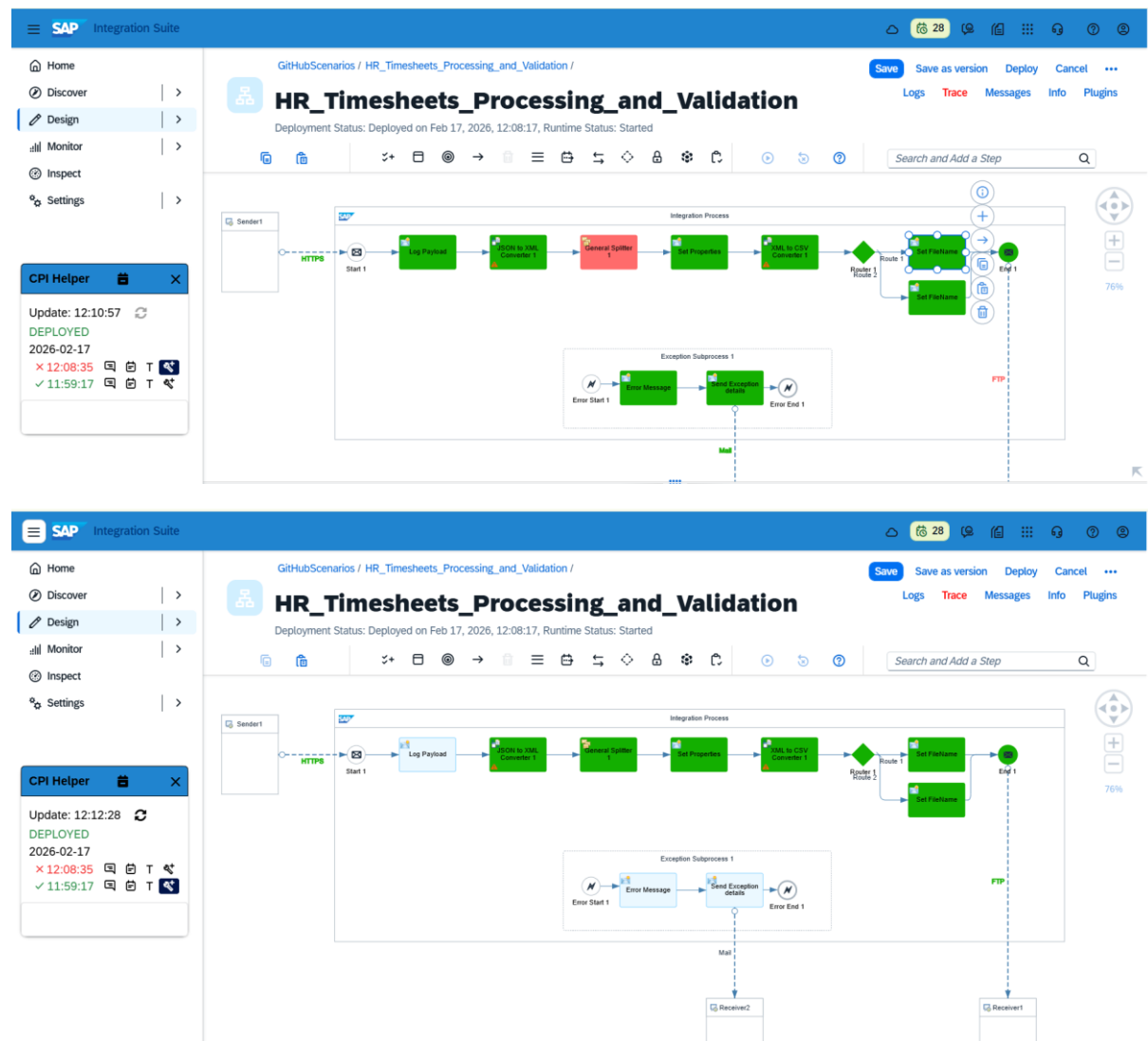
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Scenario: HR_Timesheets_Processing_and_Validation

1.Scenario Overview

This integration handles employee timesheet submissions from a web application. Timesheet records are validated, converted, and stored in the HR system in CSV format using region-specific folders. It demonstrates the use of HTTP Adapter, Splitter, Filter, Router, Format Conversion, and File Adapter, suitable for automating and validating HR operations.

2.Architechture Diagram



3. Integration Flow

Steps

1. **HTTP Sender Adapter** receives batch JSON data

HTTPS

General **Connection** Conditions

REQUEST PROCESSING

Address: /employee/timesheets

Authorization: User Role

User Role: ESBMessaging.send

CSRF Protected: ☒

2. **JSON to XML Converter** prepares structure for processing
3. **XML Splitter** breaks down entries into individual records

General Splitter

General **Processing**

Expression Type: XPath

XPath Expression: /root/Timesheets

Grouping:

Streaming: ☒

Parallel Processing: ☐

Stop on Exception: ☐

4. **Content Modifier** adds metadata (e.g., company name, processed timestamp) by **Setting Properties**

Content Modifier

General **Exchange Property** Message Header Message Body

Properties:

Action	Name	Source Type	Source Value	Data Type	Default Value
Create	EmployeeID	XPath	//EmployeeID	java.lang.String	
Create	Date	XPath	replace(//Date, '-', '')	java.lang.String	
Create	Hours	XPath	//Hours	java.lang.String	
Create	CompanyName	Constant	Infosys		
Create	ProcessedDate	Expression	\$(date:now)		

5. **Router** separates valid and invalid records using exchange property

Router

General **Processing**

ERROR HANDLING

Throw Exception: ☐

ROUTING CONDITION

Order	Route Name	Condition Expression	Default Route
1	Route 1	\$(property.Hours) <= 12	☐
2	Route 2		☒

6. **XML to CSV Converter** formats data

7. **File Receiver Adapter** writes valid CSVs to /hr/valid/, invalid ones to /hr/invalid/

The screenshot shows the configuration for an FTP File Receiver Adapter. The 'Target' tab is active, displaying the following settings:

- FILE ACCESS PARAMETERS:**
 - Directory: /hr/\${property.directory}
 - File Name: \${property.EmployeeID}_\${property.Date}.csv
 - Append Timestamp: ☐
- CONNECTION PARAMETERS:**
 - Address: ftp.drivehq.com
 - Proxy Type: Internet
 - Encryption: Explicit FTPS
 - Credential Name: FTP_Connect
 - Timeout (in ms): 10000
 - Maximum Reconnect Attempts: 3

8. **Sample Input JSON (from Web App)**

```
{
  "Timesheets": [
    { "EmployeeID": "E123", "Date": "2025-07-16", "Hours": 9 },
    { "EmployeeID": "E456", "Date": "2025-07-16", "Hours": 14 }
  ]
}
```

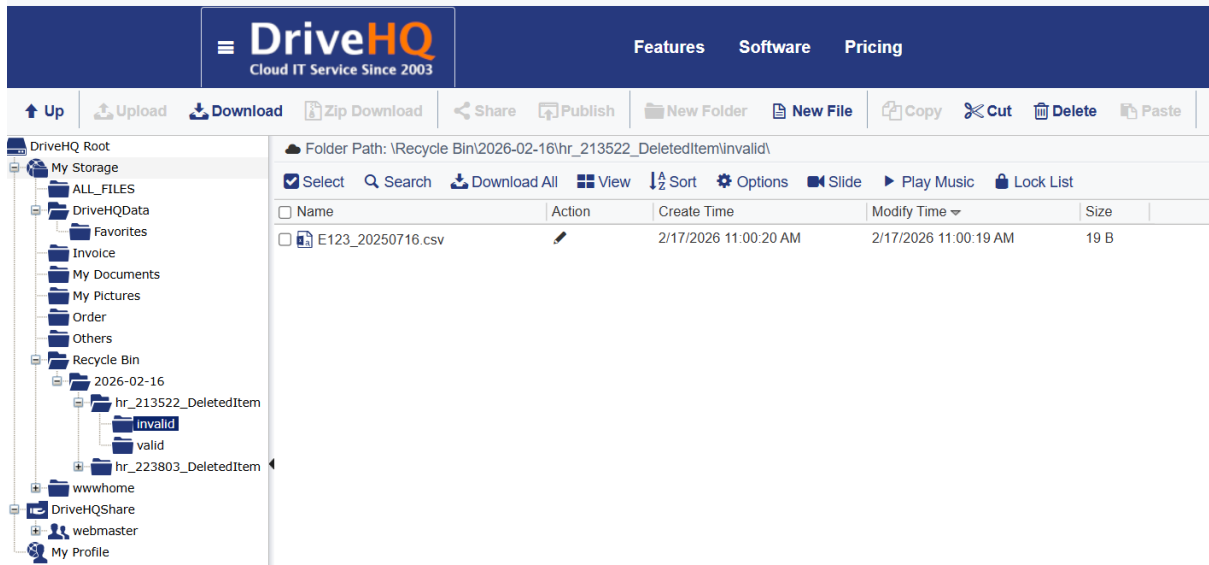
9. **Converted XML (after JSON to XML)**

```
<Timesheets>
  <Entry>
    <EmployeeID>E123</EmployeeID>
    <Date>2025-07-16</Date>
    <Hours>9</Hours>
  </Entry>
  <Entry>
    <EmployeeID>E456</EmployeeID>
    <Date>2025-07-16</Date>
```

<Hours>14</Hours>

</Entry>

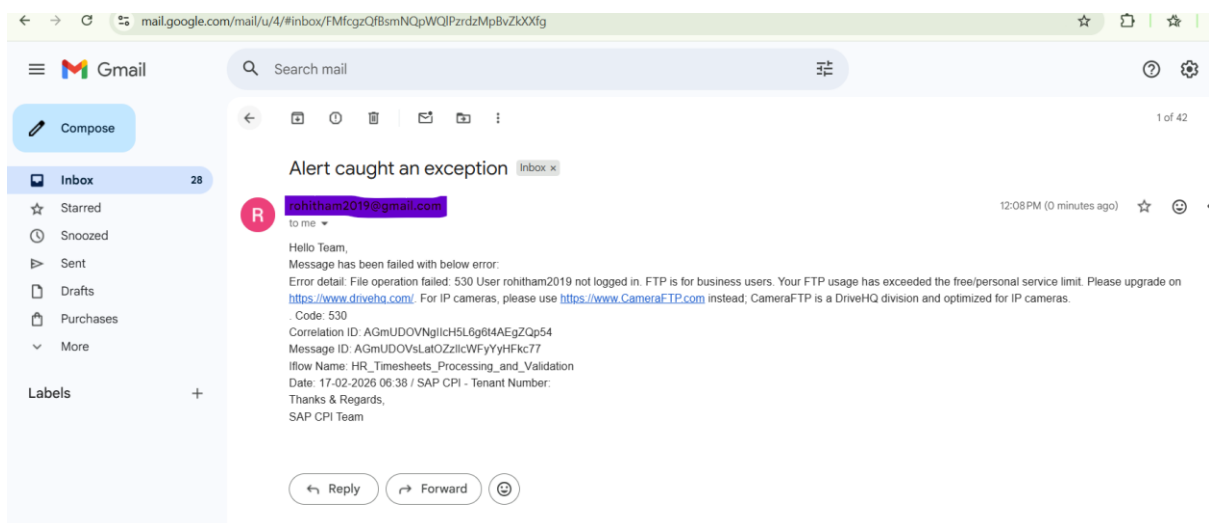
</Timesheets>



Files stored in HR system

10. Exception Handling

- Any transformation or file-write error is:
 - Alert sent via **Gmail Adapter** with employee ID and error details



11

Iflow Package



HR_Timesheets_Processing_and_Validation.zip